DECENT GROUP SPACES OVER FIELDS ARE SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Separation criterion via the unit. Groupoids in Spaces, lemma group-scheme-separated (cross-chapter, named). Reduces separation of G to closedness of the unit e : S -> G.
- Finite residue field extensions. Decent Spaces, lemma finite-residue-field-extension-finite (cross-chapter, named). Closes the unit-point argument for decent spaces.
- Diagonal of a groupoid. Groupoids in Spaces, lemma diagonal (cross-chapter, named). j = (t, s) separated once the stabilizer is separated.

Source reference. Original reference 08AL: focused-seeds/geometry/S9-smg-group-scheme-over-field-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01JO: focused-seeds/geometry/S9-smg-group-scheme-over-field-separated/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06DX: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 853–853. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06DY: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 859–859. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0455: focused-seeds/geometry/S9-smg-group-scheme-over-field-separated/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 06P6: focused-seeds/geometry/S9-smg-group-scheme-over-field-separated/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04ZH: focused-seeds/geometry/S9-smg-group-scheme-over-field-separated/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 06DY assume R is a decent space. Then R is a separated algebraic space. In Situation 06DX assume that G is a decent algebraic space. Then G is separated algebraic space.*

Proof. We first prove the second assertion. By Groupoids in Spaces, Lemma 06P6 we have to show that $e : S \to G$ is a closed immersion. This follows from Decent Spaces, Lemma 08AL.

Next, we prove the first assertion. To do this we may replace B by S. By the paragraph above the stabilizer group scheme $G \to U$ is separated. By Groupoids in Spaces, Lemma 0455 the morphism $j = (t, s) : R \to U \times_S U$ is separated. As U is the spectrum of a field the scheme $U \times_S U$ is affine (by the construction of fibre products in Schemes, Section 01JO). Hence R is separated, see Morphisms of Spaces, Lemma 04ZH. □